

Land, Credit, Overvaluation, and Debt Rollover: A Systematic Narrative Review of the Hirano-Stiglitz-Toda Branch

Review question

How does the Hirano-Stiglitz-Toda literature explain the relationship between land-price inflation, credit expansion, unbalanced growth, productive capital formation, land overvaluation, and debt rollover, and what can be carried into a Fondecyt project on Latin American land-rent financialization?

Executive synthesis

The bounded Hirano-Stiglitz-Toda branch does not treat land as a neutral background asset. Land is non-produced, durable, scarce, and able to work at once as a production factor, store of value, and collateral object. That combination creates the macro-financial problem. A rise in land prices can increase measured wealth without adding productive capacity. Credit routed toward land or real estate can enlarge balance sheets while reducing long-run growth. Differential productivity growth can pull land prices away from land rents. Once land enters the asset structure, the relation between returns, growth, and debt rollover also changes.

The core argument is not the generic claim that "financialization increases land rent." The sharper claim is that land valuation becomes macroeconomically central when credit, collateral, savings, and growth organize around a non-produced asset. Four mechanisms carry the review: land speculation crowds out capital formation; real-estate credit and collateral amplify land-price inflation; unbalanced growth can separate land prices from rent fundamentals; and land can modify R versus G and the possibility of infinite debt rollover. The mechanisms are powerful as theory and thin as institutional description. For the Fondecyt project, they must be reconstructed first on their own terms, then translated into Latin American urban land-rent and financialization contexts.

1. Introduction: the object of the review

P01. Land-price inflation poses a macroeconomic problem because it can raise balance-sheet wealth without expanding productive capacity. Land is not produced like machines, infrastructure, or productive structures. The same price increase can therefore appear as private wealth, collateral value, or capital gain while leaving the productive capital stock largely unchanged. Stiglitz's wealth measurement papers supply the conceptual base: measured wealth can rise through capitalized rents, including land and monopoly rents, even when real capital accumulation and national saving do not rise in proportion (Stiglitz, 2015c, pp. 1, 2, 4, 11; Stiglitz, 2015a, pp. 10, 11, 28).

P02. The organizing unit is the relation, not the paper. Land versus productive capital defines the object. Credit versus land price defines the financial channel. Unbalanced growth versus rent fundamentals defines the overvaluation theorem. Land, R , G , and debt rollover define the debt-sustainability channel. Wobbly dynamics remains supplementary: it explains endogenous instability and phase transitions, but it does not displace the four central mechanisms (Hirano and Stiglitz, 2022a, pp. 7, 24, 46, 47; Hirano and Stiglitz, 2022b, pp. 1, 2, 10).

P03. Reconstruction comes before critique. The Hirano-Stiglitz-Toda branch is treated first as a formal macro-financial research program with its own objects: land, land rent, land price, productive capital, wealth, collateral, leverage, R , G , and debt rollover. Its methods are overlapping-generations models, rational expectations, collateral constraints, no-arbitrage conditions, and unbalanced growth. The Fondecyt translation comes later. These models are not Latin American urban political economy, but they identify mechanisms that can travel only after their institutional assumptions are made explicit.

2. Method and corpus

P04. The corpus contains fourteen registered sources and two note layers: ninety-eight per-paper question notes and thirty-two cluster synthesis notes. Those notes function as extraction aids, not citable sources. Claims enter this draft only when they can be routed back to the registered source papers through the note system. Exact page references are kept where available. Where the notes support a claim but do not supply publication-ready citation detail, the draft marks the problem instead of inventing a page or reference.

P05. The extraction protocol keeps five dimensions separate. Definitions identify the objects: land, land rent, housing rent, real estate value, productive capital, wealth, bubble, and overvaluation. Mechanisms explain how those objects connect. Causality

orders the movement, for example from credit easing to land demand, land prices, and capital crowding-out. Variables and parameters show where the model does its work: P_t , D_t , K_t , k_{t+1} , R , G , interest rates, collateral parameters, leverage, and elasticity of substitution. Theory and evidence then separate formal results, empirical motivations, and policy implications.

P06. Cluster synthesis supplies infrastructure, not finished prose. C1 grounds the wealth-capital distinction. C2 organizes wobbly dynamics and instability. C3 gathers credit, low rates, and crowding-out. C4 links unbalanced growth, leverage, land bubbles, and overvaluation. C5 handles land, R versus G , and debt rollover. C6 is different: it is corpus-level synthesis, not a bounded cluster, so its claims require extra caution because they draw on all sources at once.

3. Conceptual foundations: land, rent, wealth, and productive capital

P07. The opening distinction separates productive capital from wealth. Productive capital consists of produced assets used in production. Wealth is wider: ownership claims over productive capital, land, capitalized rents, monopoly rents, intellectual-property rents, and other claims. That difference blocks a common inference. Rising wealth is not automatically capital deepening. In Stiglitz's formulation, the wealth residual is the gap between observed increases in wealth and the smaller increase implied by real capital accumulation and national saving (Stiglitz, 2015a, pp. 10, 11, 28; Stiglitz, 2015c, pp. 1, 2, 4, 11).

P08. Land rent and land price must also be kept apart. Land rent is a flow or dividend generated by land. Land price is the capitalized asset price of the land. They can move together, but the Land Overvaluation Theorem turns on the case where they do not. Land is overvalued when its price exceeds the present value of its rents under the relevant discounting structure. That is not the claim that every high land price is a bubble. It is a specific relation between an asset price and its rent fundamentals (Hirano and Toda, 2025b, pp. 1, 3, 4, 13).

P09. Housing and real estate introduce a composite object. Real estate joins land to produced structures. Housing rent includes the service flow from structures as well as the land component. Pure land rent is the return to the non-produced site. Toda's short paper matters because it blocks a misleading inference from observed housing rent-yield ratios. A stationary aggregate housing rent-to-price ratio does not rule out a land bubble if the analyst has not separated the return on reproducible structures from the pure land dividend (Toda, 2025, pp. 1, 2, 3).

P10. The payoff is analytical precision in the study of financialization. A bank can expand mortgage or real-estate credit, land prices can rise, and measured wealth and collateral values can jump immediately. None of that proves that the economy has produced more capital. It also does not prove that land rent has grown at the same pace as the asset price. The SHT literature uses this distinction to show how an economy can look richer on balance sheets while becoming more exposed to lower productive investment, fragile collateral values, or unstable asset-price paths (Stiglitz, 2015b, pp. 16, 17, 18; Hirano and Stiglitz, 2024, pp. 1, 15, 33, 38).

4. Mechanism I: land-price inflation and capital-formation crowding-out

P11. Crowding-out first appears when land competes with productive capital for savings. In the OLG version, young agents allocate finite savings between capital and land. A higher land price absorbs more of that savings envelope and leaves less for real capital accumulation. Hirano and Stiglitz write the relation as $k_{t+1} + P_t = s_t(w_t + e)$: higher land prices reduce capital investment in the "obvious sense" that land holdings crowd out real capital accumulation (Hirano and Stiglitz, 2022a, pp. 7, 24, 46, 47).

P12. The chain is more than accounting. A land boom raises the price of a fixed asset. Agents treat land as a store of value, and resale expectations sustain demand. Higher land prices reduce capital accumulation, lower future wages and output, and can raise the return on scarce productive capital. That higher return can require even faster land-price appreciation for arbitrage to hold. Once the path hits its bound, expectations can reverse and the boom can endogenously become a crash (Hirano and Stiglitz, 2022a, pp. 7, 24, 46, 47).

P13. The Henry George paper gives this mechanism a policy-theoretic form. Land speculation diverts scarce savings away from productive investment. A tax on land, or on returns to land speculation, can redirect resources toward productive capital and improve growth and welfare under the model's assumptions. The point is not only revenue. Land taxation changes portfolio incentives where an asset in fixed supply competes with produced capital for savings (Hirano and Stiglitz, 2025a, pp. 327, 329, 335, 345).

P14. The crowding-out mechanism is conditional. It does not say that every land purchase reduces growth. The result depends on land functioning as an alternative store of value, on savings or financing being limited at the relevant margin, on land-price

appreciation being attractive relative to productive investment, and on land purchases not creating new productive capacity. Those conditions are plausible in financialized land markets, but they must be shown empirically. For Latin American cases, this is a [source gap]: the current corpus does not itself provide evidence on the composition of credit, land tenure, informal housing production, or metropolitan land markets in the region.

5. Mechanism II: credit, collateral, leverage, and monetary policy

P15. Credit sharpens the crowding-out claim by shifting attention from quantity to allocation. Hirano and Stiglitz argue that the macroeconomic effect of credit expansion depends less on aggregate credit than on where the credit goes. Credit for productive capital can raise productivity and growth. Credit for real estate or land can fuel speculation, inflate asset prices, and reduce long-run growth. The long-run growth paper states the sectoral-credit hypothesis directly: it is "not so much aggregate credit expansion" that matters, but sectoral credit expansion (Hirano and Stiglitz, 2024, pp. 1, 15, 33, 38).

P16. The low-interest-rate paper works through the same channel. Rapid credit expansion is associated with later growth slowdowns when credit flows disproportionately into real estate, while credit expansion toward manufacturing is associated with higher productivity. The model explains why: lower rates or looser collateral constraints reduce the down payment required to acquire land, raise the leveraged return on land speculation, and move portfolios away from productive capital. What looks like easier finance in partial equilibrium can become lower productive investment in general equilibrium once land-price effects enter (Hirano and Stiglitz, 2025c, p. 1; Hirano and Stiglitz, 2024, pp. 1, 15, 33, 38).

P17. The collateral distinction carries the mechanism. Hirano and Stiglitz separate collateral constraints tied to productive capital from those tied to land. Relaxing capital collateral can crowd in productive investment. Relaxing land or real-estate collateral can crowd out productive investment if it raises the payoff to land speculation. The effect depends on the relative pledgeability of capital and land, represented in the notes through θ for capital collateral and θ^x for land collateral (Hirano and Stiglitz, 2024, pp. 4, 7, 10, 15, 33, 38; Hirano and Stiglitz, 2025c, pp. 1, 2, 16, 17, 19, 29).

P18. Stiglitz's land-and-credit analysis supplies the bridge to the later Hirano-Stiglitz models. In the Part IV paper, credit availability and bank decisions can drive land prices upward, raising measured wealth without increasing the equilibrium capital stock. Land serves as collateral, and credit conditions can inflate that collateral value. The mechanism is circular: credit supports land prices, higher land prices support collateral values, and higher collateral values support more credit. The result is balance-sheet expansion around a non-produced asset, not productive accumulation (Stiglitz, 2015b, pp. 16, 17, 18).

P19. The financial channel makes monetary policy ambiguous, not irrelevant. In a Tobin-style intuition, lower rates or higher liquidity push agents from money into productive capital. In the SHT branch, the asset menu includes land, and land can absorb liquidity that might otherwise fund capital formation. Expansionary policy can therefore support or retard growth depending on whether it relaxes productive investment finance or land-collateral finance (Hirano and Stiglitz, 2025c, pp. 1-4; Hirano and Stiglitz, 2024, pp. 1, 15, 33, 38).

6. Mechanism III: unbalanced growth and the Land Overvaluation Theorem

P20. Unbalanced growth has a narrow meaning in this branch. It refers to differential productivity growth between land and non-land factors or sectors. In the land overvaluation model, labor or human-capital productivity grows faster than land productivity. If the elasticity of substitution between land and labor exceeds one, output can keep expanding while land's productive contribution declines. Land rents grow slowly relative to wages and aggregate income, but land remains a durable store of value purchased out of rising savings (Hirano and Toda, 2025b, pp. 1, 3, 4, 13).

P21. The theorem's plain-language result is stark: land can become systematically overvalued even as its productive role declines. The notes record Theorem 1 as requiring $\sigma > 1$ and a convergent expected infinite sum involving the ratio of labor productivity to land productivity. Economically, $\sigma > 1$ makes the relevant exponent negative. Faster growth in labor productivity than land productivity suppresses land-rent growth, while rising income keeps pushing savings into land. Land prices rise faster than land rents can justify (Hirano and Toda, 2025b, pp. 4, 13).

P22. Land overvaluation is not simply irrational exuberance. It is a model result in which the price of land separates from the present value of land rents under structural growth conditions. It also differs from a generic housing-price boom. Housing combines land and structures, so an observed housing rent-yield can remain stable even if the pure land component contains a bubble. Toda's measurement warning belongs to the same architecture: the analyst must separate the rent on structures from the pure land dividend before drawing conclusions about bubbles (Toda, 2025, pp. 1, 2, 3).

P23. The leverage and asset-bubble branch extends the overvaluation logic through a different mechanism. Hirano, Jinnai, and Toda model a financial accelerator in which land prices, capital investment, land rents, wealth, and asset demand interact. Rising land prices can raise aggregate wealth and collateral values, feeding investment and asset demand in a self-reinforcing process. This is not the same as the crowding-out mechanism in the Hirano-Stiglitz credit papers. The notes indicate a channel where capital investment and land prices reinforce one another, generating a transition from fundamental equilibria to bubble equilibria under leverage and unbalanced growth (Hirano, Jinnai, and Toda, 2024, pp. 2, 3, 9).

P24. The synthesis is heterogeneous. One branch shows land speculation crowding out capital. Another shows leverage and land-price increases reinforcing capital investment up to a phase transition. That difference should remain visible. The effect of land-price inflation depends on the model architecture: whether land absorbs savings away from capital, whether collateral values relax investment constraints, whether leverage magnifies asset demand, and whether the equilibrium is fundamental or bubbly. A serious application to land-rent financialization cannot cite "the literature" as if it made one uniform claim.

7. Mechanism IV: R versus G, land, and infinite debt rollover

P25. The R-versus-G branch starts from a conventional result. In overlapping-generations economies with land, McCallum and Homburg are read as showing that a productive, non-reproducible, non-depreciating asset such as land rules out the landless Diamond-type oversaving problem. Along a balanced growth path, land rents and prices grow with output. Because land pays a positive rent, its total return exceeds the growth rate. By no arbitrage, the safe interest factor is pushed above the growth factor, $R > G$, and infinite debt rollover is ruled out (Hirano and Toda, 2025a, pp. 1, 2, 16, 17).

P26. Hirano and Toda narrow that conventional land-efficiency result. They argue that it depends on knife-edge balanced-growth restrictions, including equal productivity growth or Cobb-Douglas-like substitution conditions. Once the model allows unbalanced growth, with labor productivity growing faster than land productivity and $\sigma > 1$, land can become asymptotically negligible relative to aggregate endowments. Under those broader conditions, a Pareto-inefficient equilibrium with $R < G$ can arise even in an economy with land, reopening the possibility of infinite debt rollover (Hirano and Toda, 2025a, pp. 1, 2, 16, 17).

P27. The 2026 CIGS version cautions against using $R < G$ as a policy shortcut. The notes indicate that the paper criticizes the heuristic that market frictions generating $R < G$ automatically settle the debt-rollover question. If G , the relevant asset return, and the market frictions are not specified, the simple inequality loses theoretical coherence (Hirano and Toda, 2026, p. 4). This matters for the Fondecyt project because Latin American debt and land markets are open, institutionally heterogeneous, and exposed to currency, sovereign-risk, and external-finance conditions outside the closed OLG benchmark.

P28. The debt-rollover result should not carry more weight than the model gives it. The formal result concerns infinite debt rollover in OLG economies with land and unbalanced growth. It does not by itself theorize private developer balance sheets, mortgage refinancing, or corporate debt rollover in Latin American urban land markets. The executive memo's translation, from OLG sellers to financierized owners facing liabilities, maturities, liquidity pressure, and rollover needs, is a project-level extension rather than a result already proven in the SHT corpus. It is promising, but it must remain marked as translation rather than citation.

8. Wobbly dynamics and endogenous instability

P29. The wobbly-dynamics papers explain why land-price dynamics need not converge smoothly. In the global-dynamics branch, rational-expectations economies can fluctuate indefinitely without settling into a regular steady state. Land widens the parameter region in which such dynamics occur because it offers an alternative store of value that competes with capital and changes the relation among savings, capital accumulation, interest rates, and asset prices (Hirano and Stiglitz, 2022b, pp. 1, 2, 10; Hirano and Stiglitz, 2022a, pp. 7, 24, 46, 47).

P30. The branch contributes a theory of endogenous instability. Instability is not imposed from outside by irrational shocks. It follows from rational expectations, finite life, asset-market interactions, and phase transitions. A land-price boom can remain consistent with expectations for a time. As land absorbs savings, capital falls, wages and output fall, interest rates rise, and the required land-price appreciation becomes harder to sustain. At a threshold, the economy shifts to a crash path (Hirano and Stiglitz, 2022a, pp. 7, 24, 46, 47).

P31. Wobbly dynamics remains supplementary to the review's main argument. It gives the project a language for instability, bounded boom-bust paths, and endogenous crashes, but it does not replace the credit-allocation, overvaluation, or R-versus-G mechanisms. Its value is to prevent a static reading of land overvaluation. If land prices, credit, and capital formation interact dynamically, the question is not only whether land is overvalued at one date. The question is how valuation paths affect

investment, collateral, refinancing, and crisis timing. The specific translation to Latin American boom-bust paths remains a [source gap].

9. Policy implications inside the corpus

P32. The policy implication is not "less credit" in the abstract. Composition matters. Credit to productive capital can crowd in investment and growth; credit to real estate or land can crowd out capital and retard growth. A policy agenda drawn from the corpus would distinguish productive collateral rules from land-collateral rules, then ask whether financial liberalization raises capital formation or mainly increases leveraged land demand (Hirano and Stiglitz, 2024, pp. 1, 15, 33, 38; Hirano and Stiglitz, 2025c, p. 1).

P33. Land taxation responds directly to the speculative allocation problem. In the Henry George paper, land taxation is not only a distributional device. It changes the return to land speculation and can redirect savings toward productive capital. That policy claim remains model-dependent: it assumes that the relevant distortion is land speculation and that the state can tax land or land rents without reproducing other distortions (Hirano and Stiglitz, 2025a, pp. 327, 328, 335, 344). In Latin American settings, cadastres, informality, municipal fiscal capacity, political capture, and legal property regimes would have to be incorporated before the policy can be operationalized [source gap: Latin American institutional implementation requires external comparator literature].

P34. Monetary policy has the same compositional limit. Lower interest rates can support productive investment if the financial system channels liquidity toward capital formation. They can also raise the leveraged return to land and real estate. The SHT contribution is to show that low rates are not automatically pro-growth when land is a major store of value and collateral object. The policy implication is targeted regulation: restrict the channels that convert liquidity into land-price inflation while preserving or expanding credit for productive investment.

P35. Measurement is a policy issue as well. If wealth statistics rise because land values or capitalized rents rise, policy makers may mistake balance-sheet expansion for improved sustainability. Stiglitz's wealth measurement work warns against that interpretation. A land-rent financialization agenda should track asset prices and wealth-to-income ratios, but also productive capital formation, sectoral credit allocation, land-rent capitalization, and the separation between structures and land (Stiglitz, 2015c, pp. 1, 2, 4, 11; Toda, 2025, pp. 1, 2, 3).

10. Critical appraisal and Fondecyt translation

P36. The corpus is strongest as conceptual and mechanistic theory. It lets the project argue that land-price inflation can be macroeconomically harmful without relying on a moral claim about speculation. Land is fixed and non-produced. Land prices can capitalize rents. Land collateral can expand credit. Credit can move toward or away from productive capital. Unbalanced growth can separate land prices from rents. Debt-rollover conditions depend on the asset structure of the economy. These mechanisms give the Fondecyt project a disciplined theoretical architecture.

P37. Transportability is the weak point. The models are stylized: OLG structures, rational expectations, closed or simplified economies, representative sectors, and abstract land markets. They do not directly model Latin American urban land tenure, informal settlement, developer finance, state planning, zoning, mortgage segmentation, dollarized property markets, extractive land rents, or peripheral financial dependence. These omissions do not invalidate the mechanisms. They define the work required before those mechanisms can become claims about Latin America.

P38. The executive memo's first major translation concerns OLG. In the SHT branch, OLG solves a formal asset-circulation problem: young agents buy, old agents sell, and resale expectations can support land demand. The Fondecyt project need not adopt OLG as its ontology. It can translate the same problem into a political-economy account of owners, developers, households, or firms that sell land or real estate because they face liabilities, maturities, liquidity constraints, refinancing pressure, and balance-sheet management. The corpus has not yet established that translation. It should be developed as a project contribution and marked as such.

P39. A second translation concerns unbalanced growth. In Hirano and Toda, unbalanced growth is a formal condition about differential productivity growth and substitution between land and non-land factors. In the project's developing framework, unbalanced growth may refer to a non-proportional relation among demand, capital accumulation, and productive capacity. These are not the same concept. The review should use the SHT concept when reconstructing the theorem and reserve the project's broader disproportional-growth concept for later theoretical development. Collapsing the two would create false continuity.

P40. The corpus leaves empirical work open. It motivates questions about sectoral credit, real-estate collateral, land taxation, and land-price/rent separation, but it does not answer them for Chile or Latin America. The next empirical stage would need data on sectoral credit flows, land and structure decomposition, rent yields, collateral rules, mortgage and developer finance, property taxation, informality, and metropolitan price gradients. Until that work is done, the review can identify mechanisms and research hypotheses, not validated regional facts.

11. Conclusion

P41. The Hirano-Stiglitz-Toda branch is best read as a theory of how a non-produced asset can reorganize macro-financial relations. Land is not just another input. It is a store of value and collateral object whose price can rise without expanding productive capacity. That price increase can absorb savings, redirect credit, change collateral values, alter growth paths, and affect the relation between returns and growth. The branch gives the Fondecyt project a rigorous way to connect land rent, credit, capital formation, overvaluation, and debt rollover.

P42. The most defensible mechanisms are the wealth-capital distinction, the sectoral-credit distinction, the savings/capital crowding-out channel, the unbalanced-growth route to land overvaluation, the measurement distinction between housing rent and pure land rent, and the R-versus-G critique of conventional land-efficiency results. The fragile elements are the direct application of OLG to Latin American owners, the leap from public infinite debt rollover to private balance-sheet refinancing, and any empirical claim about regional land-rent financialization not yet supported by regional data.

P43. The next phase should keep the order disciplined: define the objects, reconstruct the mechanisms, state the causal chains, identify variables and parameters, distinguish theory from evidence, and only then translate. The review should not claim that SHT already explains Latin American land-rent financialization. It should claim that SHT provides a formal mechanism set that can be translated, tested, and modified for that project.

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Citation and source-status notes

- [source gap] appears where the review identifies a needed Latin American or institutional extension not supplied by the bounded SHT corpus.
- NotebookLM notes were not cited as sources. They were used to locate claims and page anchors in the original papers.